CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 5: **Parallel and Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The reflex angle that goes with an angle of 135° (so that together they make a complete turn) is: 1

(a) 215° **(b)** 245° **(c)** 235° **(d)** 225°

2. Two complementary angles differ by 30°. The larger angle is: 1

(a) 70° **(b)** 60° **(c)** 80° **(d)** 50°

3. Two angles of a linear pair differ by 40°. The smaller angle is: 1

(a) 70° **(b)** 90° **(c)** 60° **(d)** 80°

4. One angle of a linear pair is 3 times the other. The larger angle is: 1

(a) 135° **(b)** 145° **(c)** 155° **(d)** 125°

5. An angle is equal to its supplement. The angle is: 1

(a) 180° **(b)** 45° **(c)** 90° **(d)** 60°

6. Two supplementary angles are in the ratio $4 : 5$. The larger angle is: 1

(a) 110° **(b)** 120° **(c)** 90° **(d)** 100°

7. The supplement of the complement of 50° is: 1

(a) 150° **(b)** 140° **(c)** 130° **(d)** 160°

8. The reflex angle that goes with an angle of 100° (so that together they make a complete turn) is: 1

(a) 270° **(b)** 260° **(c)** 280° **(d)** 250°

9. **Assertion (A):** Vertically opposite angles are equal. 1

Reason (R): Vertically opposite angles always add up to 180°.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** When a transversal cuts two parallel lines, the corresponding angles are supplementary. 1

Reason (R): When a transversal cuts two parallel lines, the corresponding angles are equal.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. An angle is 10° more than 3 times its complement. Find the angle. 2

12. Two supplementary angles differ by 34°. Find both angles and the complement of the smaller one. 2

13. Find the angle whose supplement is 6 times its complement. 2

14. An angle is 14° more than 3 times its complement. Find the angle. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Angles on one side of a straight line at a point are 40°, 60°, 25° and a fourth angle. Find the fourth angle and name the kind of each angle. 3

16. Two supplementary angles are in the ratio $1 : 9$. Find the complement of the smaller angle and the supplement of the larger angle. 3

17. Two adjacent angles form a linear pair and one of them is 100°. Find the other angle, the half of each angle and the angle between the two bisectors. 3

OR

Angles on one side of a straight line at a point are 15°, 85°, 45° and a fourth angle. Find the fourth angle and name the kind of each angle. 3

18. Two supplementary angles are in the ratio $2 : 13$. Find the complement of the smaller angle and the supplement of the larger angle. 3

OR

Two adjacent angles form a linear pair and one of them is 110°. Find the other angle, the half of each angle and the angle between the two bisectors. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Two roads cross at a point O, forming four angles. One of the angles is 40°. 5

(i) What is the angle vertically opposite to it?

(ii) What is the angle adjacent to it?

(iii) What is the sum of the four angles?

(iv) In another crossing, two adjacent angles are $(2x + 10)^\circ$ and $(3x - 20)^\circ$. Find x and the two angles.

20. Three angles on one side of a straight line at a point are in the ratio $2 : 3 : 7$. Find the three angles and name the kind of each. 5

OR

A transversal cuts two parallel lines and two interior angles on the same side are in the ratio $1 : 11$. Find these two angles and say how many of the eight angles measure 15° and how many 165°. 5

END OF QUESTION PAPER — Total: 40 Marks